

AKTU B.Tech 2nd Year Syllabus — Easy-to-Read RAG Version

Designed so that a person can read it aloud naturally and a retrieval system can clearly separate subject, unit, and listed syllabus material.

How this syllabus is organized

How to read this document: A subject is followed by its units. Each unit is followed by the material taught in that unit.

When a student says 'topics in Unit 1', 'what is in Unit 1', 'what do we study in Unit 1', or 'what is covered in Unit 1', they are asking for the material listed under that unit.

When a student asks 'subjects in Semester III', use the Semester III subject list near the beginning.

Do not treat this guide as additional academic content. It only explains how to interpret the syllabus structure.

Semester III — Subjects

BAS303 / BOE3**

- Mathematics-III / Mathematics-IV / Mathematics-V (science based/open elective, depending on branch)

BVE301 / BAS301

- Universal Human Value and Professional Ethics / Technical Communication

BCS301

- Data Structure

BCS302

- Computer Organization and Architecture

BCS303

- Discrete Structures and Theory of Logic

BCS351

- Data Structure Lab

BCS352

- Computer Organization and Architecture Lab

BCS353

- Web Designing Workshop

BCC301

- Cyber Security / Python

BCC302

- Programming Internship Assessment

BCC351

- Mini Project (assessment only)

Semester IV — Subjects

BAS403 / BOE4**

- Mathematics-III / Mathematics-IV / Mathematics-V / Science Based Open Elective

BAS401 / BVE401

- Technical Communication / Universal Human Value and Professional Ethics

BCS401

- Operating System

BCS402

- Theory of Automata and Formal Languages

BCS403

- Object Oriented Programming with Java

BCS451

- Operating System Lab

BCS452

- Object Oriented Programming with Java Lab

BCS453

- Cyber Security Workshop

BCC402 / BCC401

- Python Programming / Cyber Security

BVE451 / BVE452

- Sports and Yoga-II / NSS-II

BCS303 — Discrete Structures and Theory of Logic

Unit 1

- Set Theory and Relations: introduction to sets and combination of sets.
- Relations: definition, operations on relations, properties of relations, composite relations, equality of relations, recursive definition of relation, and order of relations.
- POSET and Lattices: Hasse Diagram, POSET, and the definition and properties of lattices.

- Types of lattices listed in the syllabus: bounded, complemented, distributed, modular, and complete lattice.

Unit 2

- Functions: definition, classification of functions, and operations on functions.
- Growth of functions.
- Boolean Algebra: introduction, axioms and theorems of Boolean algebra, algebraic manipulation of Boolean expressions, simplification of Boolean functions, and Karnaugh maps.

Unit 3

- Theory of Logics: proposition, truth tables, tautology, satisfiability, contradiction, algebra of proposition, and theory of inference.
- Predicate Logic: first-order predicate, well-formed formula of predicate, quantifiers, and inference theory of predicate logic.

Unit 4

- Algebraic Structures: definition, groups, subgroups and order, cyclic groups, cosets, Lagrange's theorem, normal subgroups, permutation and symmetric groups, group homomorphisms, and the definition and elementary properties of rings and fields.

Unit 5

- The supplied source text continues with the graph-theory portion of the Discrete Structures syllabus. Use the official source document for the exact wording when a question requires details beyond the extracted unit text.

BCS301 — Data Structure

Unit 1

- Introduction: basic terminology, elementary data organization, built-in data types in C, algorithm, efficiency of an algorithm, time and space complexity, asymptotic notations Big O, Big Theta and Big Omega, time-space trade-off, and Abstract Data Types (ADT).
- Arrays: definition, single and multidimensional arrays, representation of arrays using row-major and column-major order, derivation of index formulae for 1-D, 2-D, 3-D and n-D arrays, applications of arrays, and sparse matrices and their representations.

BCS302 — Computer Organization and Architecture

Unit 1

- The supplied source contains the detailed COA syllabus, including memory organization, cache memories, address mapping and replacement, auxiliary memories, virtual memory, input/output, interrupts, modes of data transfer, I/O channels and processors, and serial communication.
- Because the original PDF uses tables, the wording above is kept close to the extracted source rather than inventing a new syllabus.

BCS401 — Operating System

Unit 1

- Introduction: operating system and functions.
- Classification of operating systems: Batch, Interactive, Time Sharing, Real Time System, Multiprocessor Systems, Multiuser Systems, Multiprocess Systems, and Multithreaded Systems.
- Operating System Structure: layered structure, system components, operating system services, reentrant kernels, monolithic systems, and microkernel systems.

Unit 2

- Concurrent Processes: process concept, principle of concurrency, producer/consumer problem, mutual exclusion, critical section problem, Dekker's solution, Peterson's solution, semaphores, Test and Set operation.
- Classical concurrency problems: Dining Philosopher Problem and Sleeping Barber Problem.
- Inter Process Communication models and schemes, and process generation.

Unit 3

- CPU Scheduling: scheduling concepts, performance criteria, process states, process transition diagram, schedulers, Process Control Block (PCB), process address space, process identification information, threads and their management, scheduling algorithms, and multiprocessor scheduling.
- Deadlock: system model, deadlock characterization, prevention, avoidance and detection, and recovery from deadlock.

Unit 4

- Memory Management: basic bare machine, resident monitor, multiprogramming with fixed partitions, multiprogramming with variable partitions, protection schemes, paging, segmentation, paged segmentation, virtual memory concepts, demand paging, performance of demand paging, page replacement algorithms, thrashing, cache memory organization, and locality of reference.

Unit 5

- I/O Management and Disk Scheduling: I/O devices and I/O subsystems, I/O buffering, disk storage and disk scheduling, RAID.
- File System: file concept, file organization and access mechanism, file directories, file sharing, file system implementation issues, file system protection and security.

BCS402 — Theory of Automata and Formal Languages

Unit 1

- Basic Concepts and Automata Theory: introduction to Theory of Computation, automata, computability and complexity, alphabet, symbol, string, formal languages.
- Deterministic Finite Automaton (DFA): definition, representation, acceptability of a string and language.
- Non Deterministic Finite Automaton (NFA), equivalence of DFA and NFA, NFA with epsilon-transition, equivalence of NFA with and without epsilon-transition.
- Finite Automata with output: Moore Machine, Mealy Machine, equivalence of Moore and Mealy Machine, and minimization of finite automata.

Unit 2

- Regular Expressions and Languages: regular expressions, transition graph, Kleene's Theorem, finite automata and regular expression, Arden's theorem, algebraic method using Arden's theorem.
- Regular and non-regular languages: closure properties of regular languages, Pigeonhole Principle, Pumping Lemma, application of Pumping Lemma, decidability and decision properties.

Unit 3

- Regular and Non-Regular Grammars: Context Free Grammar (CFG), definition, derivations, languages, derivation trees and ambiguity.
- Regular grammars: right linear and left linear grammars, conversion of FA into CFG and regular grammar into FA, simplification of CFG, Chomsky Normal Form (CNF), Greibach Normal Form (GNF), Chomsky Hierarchy, and programming problems based on properties of CFGs.

Unit 4

- Push Down Automata and Properties of Context Free Languages: Nondeterministic Pushdown Automata (NPDA), definition, moves, language accepted by NPDA.
- Deterministic Pushdown Automata (DPDA), deterministic context-free languages (DCFL), pushdown automata for context-free languages, context-free grammars for pushdown automata, two-stack pushdown automata, Pumping Lemma for CFL, closure properties of CFL, decision problems of CFL, and programming problems based on properties of CFLs.

Unit 5

- Turing Machines and Recursive Function Theory: basic Turing Machine model, representation of Turing Machines, language acceptability, techniques for Turing Machine construction, modifications of Turing Machine.
- Turing Machine as computer of integer functions, Universal Turing Machine, Linear Bounded Automata, Church's Thesis, recursive and recursively enumerable language, Halting Problem, Post's Correspondance Problem, and introduction to Recursive Function Theory.

BCS403 — Object Oriented Programming with Java

Reference point

- The supplied source includes Java/OOP material and an indicative lab list. The lab list includes Java basics, OOP concepts, inheritance, polymorphism, exception handling, multithreading, packages, Java I/O, Spring Framework, RESTful web services using Spring Boot, and frontend web application testing with Spring Boot.

Source and scope

The academic content in this guide is based on the supplied AKTU B.Tech 2nd Year Computer Science and Engineering syllabus text.

Some parts of the supplied source are table-based. Where extraction is incomplete, this guide deliberately does not invent missing details.